

WEEK-13

PROGRAM 1

Given an array of numbers, find the index of the smallest array element (the pivot), for which the sums of all elements to the left and to the right are equal. The array may not be reordered.

Example

`arr=[1,2,3,4,6]`

- the sum of the first three elements, $1+2+3=6$. The value of the last element is 6.
- Using zero based indexing, `arr[3]=4` is the pivot between the two subarrays.
- The index of the pivot is 3.

Function Description

Complete the function `balancedSum` in the editor below.

`balancedSum` has the following parameter(s):

`int arr[n]`: an array of integers

Returns:

`int`: an integer representing the index of the pivot

Constraints

- $3 \leq n \leq 10^5$
- $1 \leq \text{arr}[i] \leq 2 \times 10^4$, where $0 \leq i < n$
- It is guaranteed that a solution always exists.

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n , the size of the array arr .

Each of the next n lines contains an integer, $\text{arr}[i]$, where $0 \leq i < n$.

Sample Case 0

Sample Input 0

STDIN Function Parameters

4 → $\text{arr}[]$ size $n = 4$

1 → $\text{arr} = [1, 2, 3, 3]$

2

3

3

Sample Output 0

2

Explanation 0

- The sum of the first two elements, $1+2=3$. The value of the last element is 3.
- Using zero based indexing, $\text{arr}[2]=3$ is the pivot between the two subarrays.
- The index of the pivot is 2.

Sample Case 1

Sample Input 1

STDIN Function Parameters

3 → arr[] size n = 3

1 → arr = [1, 2, 1]

2

1

Sample Output 1

1

Explanation 1

- The first and last elements are equal to 1.
- Using zero based indexing, $\text{arr}[1]=2$ is the pivot between the two subarrays.
- The index of the pivot is 1.

CODE

REC-CIS

```
1 /*
2  * Complete the 'balancedSum' function below.
3  *
4  * The function is expected to return an INTEGER.
5  * The function accepts INTEGER_ARRAY arr as parameter.
6  */
7
8 int balancedSum(int arr_count, int* arr)
9 {
10     int left = 0, right = 0;
11     for(int i=0;i<arr_count;i++){
12         right += arr[i];
13     }
14     for(int i=0;i<arr_count;i++){
15         if(left==(right-arr[i]))
16             return i;
17         left += arr[i];
18         right -= arr[i];
19     }
20     return 1;
21 }
22
23
```

OUTPUT

	Test	Expected	Got	
✓	int arr[] = {1,2,3,3}; printf("%d", balancedSum(4, arr))	2	2	✓

Passed all tests! ✓

PROGRAM 2

Calculate the sum of an array of integers.

Example

```
numbers = [3, 13, 4, 11, 9]
```

The sum is $3 + 13 + 4 + 11 + 9 = 40$.

Function Description

Complete the function `arraySum` in the editor below.

`arraySum` has the following parameter(s):

`int numbers[n]`: an array of integers

Returns

`int`: integer sum of the numbers array

Constraints

$1 \leq n \leq 10^4$

$1 \leq \text{numbers}[i] \leq 10^4$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n , the size of the array numbers.

Each of the next n lines contains an integer $\text{numbers}[i]$ where $0 \leq i < n$.

Sample Case 0

Sample Input 0

STDIN	Function
-------	----------

-----	-----
-------	-------

5	→ numbers[] size $n = 5$
---	--------------------------

1	→ numbers = [1, 2, 3, 4, 5]
---	-----------------------------

2	
---	--

3	
---	--

4	
---	--

5	
---	--

Sample Output 0

15

Explanation 0

$$1 + 2 + 3 + 4 + 5 = 15.$$

Sample Case 1

Sample Input 1

STDIN	Function
-------	----------

-----	-----
-------	-------

2	→ numbers[] size n = 2
---	------------------------

12	→ numbers = [12, 12]
----	----------------------

12	
----	--

Sample Output 1

24

Explanation 1

$$12 + 12 = 24.$$

CODE

REC-CIS

```
1 /*
2  * Complete the 'arraySum' function below.
3  *
4  * The function is expected to return an INTEGER.
5  * The function accepts INTEGER_ARRAY numbers as parameter.
6  */
7
8 int arraySum(int numbers_count, int *numbers)
9 {
10     int sum=0;
11     for(int i=0;i < numbers_count;i++){
12         sum += numbers[i];
13     }
14     return sum;
15 }
16
17
```

OUTPUT

	Test	Expected	Got	
✓	int arr[] = {1,2,3,4,5}; printf("%d", arraySum(5, arr))	15	15	✓

Passed all tests! ✓

PROGRAM 3

Given an array of n integers, rearrange them so that the sum of the absolute differences of all adjacent elements is minimized. Then, compute the sum of those absolute differences. Example $n = 5$ $arr = [1, 3, 3, 2, 4]$ If the list is rearranged as $arr' = [1, 2, 3, 3, 4]$, the absolute differences are $|1 - 2| = 1$, $|2 - 3| = 1$, $|3 - 3| = 0$, $|3 - 4| = 1$. The sum of those differences is $1 + 1 + 0 + 1 = 3$. Function Description Complete the function `minDiff` in the editor below. `minDiff` has the following parameter: `arr`: an integer array Returns: `int`: the sum of the absolute differences of adjacent elements Constraints $2 \leq n \leq 105$ $0 \leq arr[i] \leq 109$, where $0 \leq i < n$ Input Format For Custom Testing The first line of input contains an integer, n , the size of `arr`. Each of the following n lines contains an integer that describes `arr[i]` (where $0 \leq i < n$). Sample Case 0 Sample Input For Custom Testing STDIN Function ----- 5 $\rightarrow$ `arr[]` size $n = 5$ 5 $\rightarrow$ `arr[]` = [5, 1, 3, 7, 3] 1 3 7 3 Sample Output 6 Explanation $n = 5$ $arr = [5, 1, 3, 7, 3]$ If `arr` is rearranged as `arr' = [1, 3, 3, 5, 7]`, the differences are minimized. The final answer is $|1 - 3| + |3 - 3| + |3 - 5| + |5 - 7| = 6$. Sample Case 1 Sample Input For Custom Testing STDIN Function ----- 2 $\rightarrow$ `arr[]` size $n = 2$ 3 $\rightarrow$ `arr[]` = [3, 2] 2 Sample Output 1 Explanation $n = 2$ $arr = [3, 2]$ There is no need to rearrange because there are only two elements. The final answer is $|3 - 2| = 1$.

CODE

 Week-13-Passing Arrays and Strings to Functions: Attempt review | REC-CIS - Google Chrome

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REC-CIS

```
1  /*
2   * Complete the 'minDiff' function below.
3   *
4   * The function is expected to return an INTEGER.
5   * The function accepts INTEGER_ARRAY arr as parameter.
6   */
7
8  int minDiff(int arr_count, int* arr)
9  {
10     for(int i=0; i<arr_count-1;i++){
11         for(int j=0;j<arr_count-i-1;j++){
12             if(arr[j]>arr[j+1]){
13                 int temp = arr[j];
14                 arr[j]=arr[j+1];
15                 arr[j+1]=temp;
16             }
17         }
18     }
19     int sum=0;
20     for(int i=0;i<arr_count-1;i++){
21         sum += abs(arr[i]-arr[i+1]);
22     }
23     return sum;
24 }
25
```

OUTPUT

	Test	Expected	Got	
✓	<pre>int arr[] = {5, 1, 3, 7, 3}; printf("%d", minDiff(5, arr))</pre>	6	6	✓

Passed all tests! ✓